

EEE5544 Stochastic Methods for Engineering 1

TAN F. WONG

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0.1 Introduction

0.1.1 What is *random*?

- Some behavior that we cannot perfectly predict, as opposed to being *deterministic*.

0.1.2 Why random?

- **Theologian:** We can't know God's mind!
- **Copenhagen quantum physicist:** The laws of quantum physics are fundamentally random.

0.1.3 What to do with random?

- Although we cannot perfectly determine random behavior, we still want to obtain its specifications to some degree.
- **Probability Theory** is the theory that systematically studies and characterizes randomness.
- In this course, we will take an engineering approach to Probability Theory—more operational and less concerned with mathematical rigor.

0.1.4 How to study probability?

1. Intuition

- **Example:** Dr. Wong is a nice guy; he will probably give me a good grade in EEE 5544.
- This approach is not systematic, relies on guessing, and cannot quantify the “*likelihood*” of the behavior.

2. Relative frequency (a.k.a. counting)

- **Example:** Dr. Wong taught EEE 5544 three times in the past. Altogether, 200 students took Dr. Wong's classes, and 50 of them received an *A*. Thus, the chance of receiving an *A* in Dr. Wong's class is

$$\frac{50}{200} = \frac{1}{4}. \quad (1)$$

- This approach assumes that every grade is “equally likely,” which may not be the case.
- What if there are infinitely many possibilities?

3. Axiomatic theory

- Developed by Kolmogorov in the early 1900s.
- Identify a few basic axioms that fit intuition.
- Build a mathematically consistent system based on the axioms.
- Use measure theory to construct the mathematical system.
- This is the approach that we will follow.

0.2 Probability Space

0.2.1 Random Experiment

- A **random experiment** is an experiment in which the outcome varies in an unpredictable fashion when the experiment is repeated under the same conditions, i.e., *random*.
- An **outcome** of a random experiment is a result that cannot be decomposed into other results.

0.2.2 Sample Space

- The set of all possible outcomes for a random experiment is called the **sample space** Ω .

Example 0.2.1. 1. Tossing a coin: $\Omega = \{H, T\}$

2. Tossing a coin twice: $\Omega = \{HH, HT, TH, TT\}$

3. Taking EEE 5544 and getting a grade: $\Omega = \{A, A-, B+, B, B-, C+, C, D, F\}$

4. Picking an integer at random: $\Omega = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

5. Checking the temperature (in K) at a random location: $\Omega = (0, \infty)$

- **Finite** Ω : Examples 1, 2, and 3
- **Countably infinite** Ω : Example 4
- **Discrete** Ω : Examples 1–4
- **Continuous (uncountably infinite)** Ω : Example 5

0.2.3 Event Class

- An **event** is a set of outcomes, i.e., a subset of Ω , whose “likelihood” can be quantified.
- Two special events:
 - $\Omega = \text{certain event}$
 - $\emptyset = \text{impossible event}$
- An **event class** $\mathcal{F}$ of a random experiment is a collection of events.

Discrete Sample Space

For a discrete Ω , we typically consider the event class consisting of all subsets of Ω , i.e., $\mathcal{F} = 2^\Omega$. This is called the *power set* of Ω .

Example 0.2.2. 1. Tossing a coin: $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$.

2. Tossing a coin twice:

$$\mathcal{F} = \{\emptyset, \{HH\}, \{TT\}, \{HT\}, \{TH\}, \{HH, TT\}, \{HH, HT\}, \dots, \{HH, TT, HT\}, \dots, \{HH, TT, HT, TH\}\}.$$

3. If $\Omega = \{a_1, a_2, \dots, a_N\}$, then

$$\mathcal{F} = \{\emptyset, \{a_1\}, \dots, \{a_N\}, \{a_1, a_2\}, \dots, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_N\}\}.$$

Altogether, there are 2^N events (why?).

4. For a countably infinite sample space $\Omega = \{a_1, a_2, \dots\}$, we can use the same idea as in the finite case, using the power set as the event space:

$$\mathcal{F} = 2^\Omega = \{\emptyset, \{a_1\}, \dots, \{a_N\}, \{a_1, a_2\}, \dots, \{a_1, a_2, a_3\}, \dots, \{a_1, a_2, \dots, a_n\}, \dots\}.$$

This enumerates all subsets of Ω . However, we need a better way to describe the power set.

Continuous Sample Space

- For a continuous sample space, such as $\Omega = (0, \infty)$, there are uncountably infinitely many outcomes. Therefore, the enumeration method used in the finite case cannot be used to construct the event class.
- To discuss the solution, we first need to study set operations.

0.2.4 Set Operations

- **Union:** $A \cup B = \{x \in \Omega : x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x \in \Omega : x \in A \text{ and } x \in B\}$
- **Complement:** $A^c = \{x \in \Omega : x \notin A\}$
- **Difference:** $A \setminus B = \{x \in \Omega : x \in A \text{ and } x \notin B\} = A \cap B^c$

Remark 0.2.1 (De Morgans Laws). 1. $(A \cup B)^c = A^c \cap B^c$.

2. $(A \cap B)^c = A^c \cup B^c$.

Proof. We prove the first identity as an example.

Since

$$\begin{aligned} x \in (A \cup B)^c &\Rightarrow x \notin A \cup B \\ &\Rightarrow x \notin A \text{ and } x \notin B \\ &\Rightarrow x \in A^c \text{ and } x \in B^c \\ &\Rightarrow x \in A^c \cap B^c, \end{aligned}$$

we have $(A \cup B)^c \subseteq A^c \cap B^c$.

Conversely, since

$$\begin{aligned} x \in A^c \cap B^c &\Rightarrow x \in A^c \text{ and } x \in B^c \\ &\Rightarrow x \notin A \text{ and } x \notin B \\ &\Rightarrow x \notin A \cup B \\ &\Rightarrow x \in (A \cup B)^c, \end{aligned}$$

we have $A^c \cap B^c \subseteq (A \cup B)^c$.

Combining the two inclusions gives $(A \cup B)^c = A^c \cap B^c$.

□

Remark 0.2.2. *Union and complement are enough to specify all other set operations.*

0.2.5 Field (Algebra)

- A **field (algebra)** is a collection of subsets of Ω that is closed under union and complement and contains both $\emptyset$ and Ω . Thus, $\mathcal{F}$ is a field if it satisfies the following conditions:
 1. $\emptyset \in \mathcal{F}$ and $\Omega \in \mathcal{F}$.
 2. If $A \in \mathcal{F}$ and $B \in \mathcal{F}$, then $A \cup B \in \mathcal{F}$.
 3. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
- Operationally, a field contains all subsets of Ω that can be obtained by applying a finite number of set operations.

Remark 0.2.3. *For any finite Ω , the power set 2^Ω is a field.*

0.2.6 σ -Field (σ -Algebra)

- A field $\mathcal{F}$ is a **σ -field** if it satisfies the additional condition
 4. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.
- Operationally, a σ -algebra contains all subsets of Ω that can be obtained through a countable number of set operations.

Remark 0.2.4. For countably infinite Ω , 2^{Ω} is a σ -field.

Generating a σ -Field

- For a continuous (uncountably infinite) sample space Ω , we often *generate* σ -fields from collections of subsets of Ω .
- For a collection $\mathcal{A}$ of subsets of Ω , consider the σ -field generated by $\mathcal{A}$: the smallest σ -field containing every subset that can be formed by performing countable set operations on members of $\mathcal{A}$.
- The σ -field generated by $\mathcal{A}$ is denoted by $\sigma(\mathcal{A})$.

0.2.7 Event Class for a Continuous Sample Space

- For a continuous (uncountably infinite) sample space Ω , we consider event classes that are σ -fields on Ω .

Example 0.2.3 (Event class for $\Omega = \mathbb{R}$)

- Consider the collection $\mathcal{A}$ of all subsets of $\mathbb{R}$ of the form $(-\infty, a)$ for some $a \in \mathbb{R}$.
- The σ -field generated by $\mathcal{A}$ is called the *Borel σ -field*. It is commonly used as an event class for the sample space $\mathbb{R}$.
- The Borel σ -field contains essentially all events of practical interest.
- Although the Borel sets form a very rich class of subsets of $\mathbb{R}$, some subsets of $\mathbb{R}$ are not Borel sets. This is critically important for allowing the likelihoods of events to be quantified.

0.2.8 Description of a Random Experiment

- The sample space Ω and an event class $\mathcal{F}$ can be used to describe a random experiment mathematically.
- We also need a way to quantify the likelihood of events. This is done by a mapping called a *probability measure* P , which will be discussed in the next section.
- In summary, the mathematical object describing a random experiment is the triple $(\Omega, \mathcal{F}, P)$, called the ***probability space*** of the random experiment.